SYNTHETIC SAMPLE PAPER - fictional, for pipeline testing only

Comparing Transformer and mBART for Low-Resource Machine Translation

Abstract

We study low-resource machine translation for Swahili and Yoruba. We compare a Transformer baseline with mBART fine-tuning on the FLORES-200 dataset.

1 Introduction

Low-resource machine translation remains difficult because parallel data is scarce. Large multilingual pretrained models such as mBART may transfer knowledge to languages with little data.

2 Method

The Transformer baseline is trained from scratch on the available parallel data. mBART is fine-tuned on the same parallel data for each language.

3 Results

On Swahili, mBART reaches 22.4 BLEU on FLORES-200, while the Transformer baseline reaches 17.9 BLEU. On Yoruba, mBART reaches 14.2 BLEU on FLORES-200, while the Transformer baseline reaches 10.8 BLEU.

4 Limitations

We report only BLEU, which may not capture morphological errors.

5 Future Work

In future work we plan to study dialectal varieties such as Algerian Arabic.